

CENTRIFUGAL PUMP**Introduction**

Centrifugal pumps are classified as rotodynamic type of pumps in which dynamic pressure is developed which enables the lifting of liquids from a lower to a higher level. The basic principle on which a centrifugal works is that when a certain mass of liquid is made to rotate by an external force, it is thrown away from the central axis of rotation and a centrifugal head is impressed which enable it to rise to a higher level. Now, if more liquid is constantly made available at the centre of rotation, a continuous supply of liquid at a higher level may be ensured. Since in these pumps the lifting of the liquid is due to centrifugal action, these pumps are called 'centrifugal pumps'

Advantages of centrifugal pumps over reciprocating pumps

The main advantage of a centrifugal pump is that its discharging capacity is very much greater than a reciprocating pump which can handle relatively small quantity of liquid only. A centrifugal pump can be operated at very high speeds without any danger of separation and cavitation. The maintenance cost of a centrifugal pump is low and only periodical check up is sufficient. But for a reciprocating pump the maintenance cost is high because the parts such as valves etc., may need frequent replacement.

Centrifugal pump

The centrifugal pump acts as a reversed of an inward radial flow reaction turbine. This means that the flow in centrifugal pumps is in the radial outward directions. The centrifugal pump works on the principle of forced vortex flow which means that when a certain mass of liquid is rotated by an external torque, the rise is pressure head of the rotation liquid takes place. The rise in pressure head at any point of the rotating liquid is proportional to the square of tangential velocity of the liquid at that point

$$(i.e., \text{rise in pressure head} = v^2/2g \text{ or } \omega^2 r^2 / 2g)$$

This at the outlet of the impeller where radius is more, the rise in pressure head will be more and the liquid will be discharged at the outlet with a high pressure head. Due to this high pressure head, the liquid can be lifted to a high level.

Classification of Centrifugal Pumps

- a) Single stage
- b) Multi stage

Component Parts of a Centrifugal Pump

The main component parts of a centrifugal pump are:

- impeller
- casing
- suction pipe
- delivery pipe

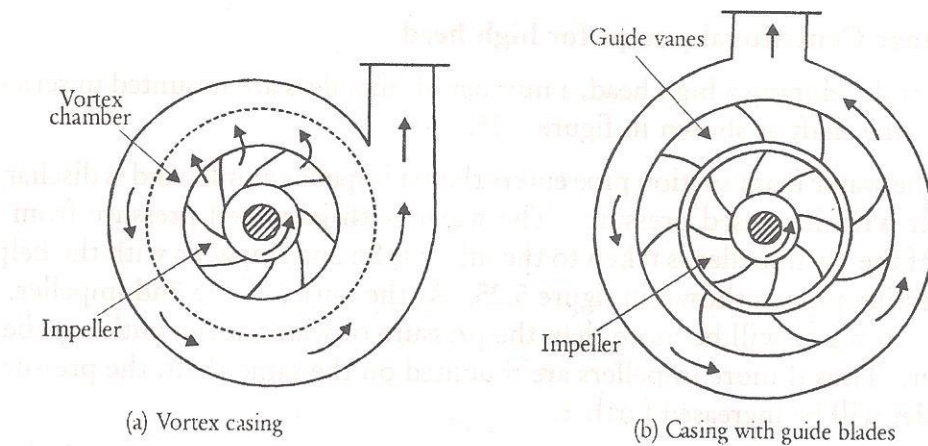
1. Impeller

The rotating part of a centrifugal pump is called 'impeller'. It consists of a series of backward curved vanes. The impeller is mounted on a shaft which is connected to the shaft of an electric motor.

2. Casing

The casing of a centrifugal pump is similar to the casing of a reaction turbine. It is an air – tight passage surrounding the impeller and is designed in such a way that the kinetic energy of the water discharged at the outlet of the impeller is converted into pressure energy before the water leaves the casing and enters the delivery pipe. The following three types of the casings are commonly adopted:

- a) Volute casing
- b) Vortex casing
- c) Casing with guide blades.

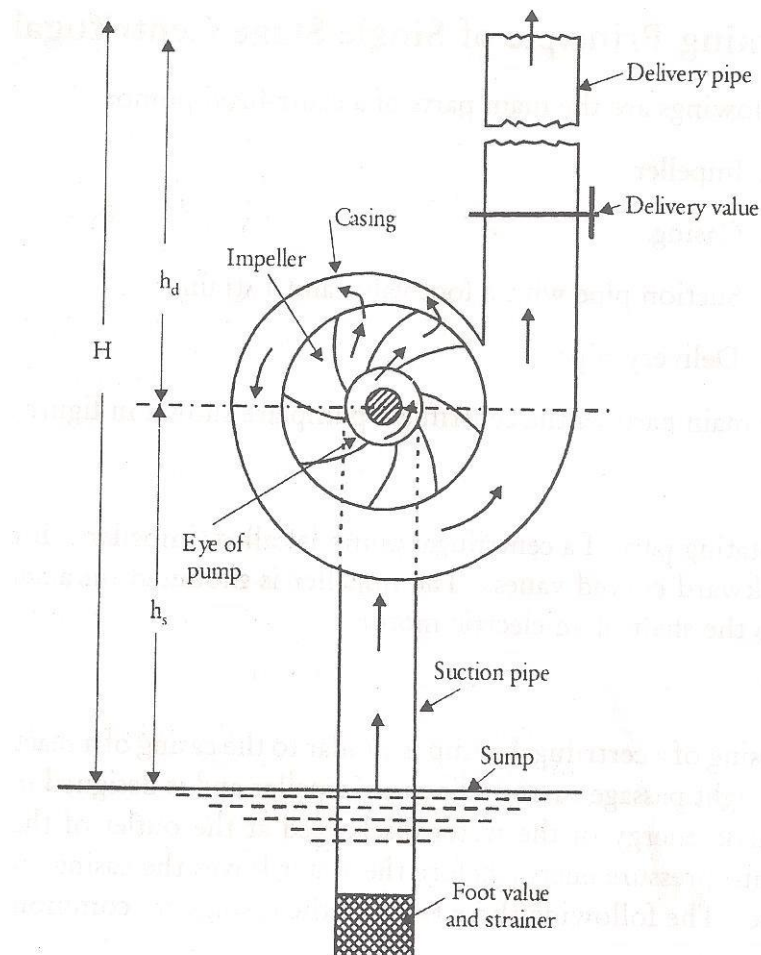


a) Volute casing

Shows the volute casing which surrounds the impeller. It is of spiral type in which area of flow increases gradually. The increase in area of flow decreases the velocity of flow. The decrease in velocity increases the pressure of the water flowing through the casing. It has been observed that in case of volute casing, the efficiency of the pump increases slightly as a large amount of energy is lost due to the formation of eddies in this type of casing.

b) Vortex casing

If a circular chamber is introduced between the casing and the impeller, the casing is known as Vortex casing. By introducing the circular chamber, the loss of energy due to the formation of eddies is reduced to a considerable extent. Thus the efficiency of the pump is more than the efficiency when only volute casing is provided.



c) Casing with guide blades

This casing is in which the impeller is surrounded by a series of guide blades mounted on a ring which is known as diffuser. The guide vanes are designed in such a way that the water from the impeller enters the guide vanes without shock. Also the area of the guide vanes increases, thus reducing the velocity of flow through guide vanes and consequently increasing the pressure of water. The water from the guide vanes then passes through the surrounding casing which is in most of the cases concentric with the impeller.

3. Suction pipe with a foot – valve and a strainer

A pipe whose one end is connected to the inlet of the pump and other end dips into water in a sump is known as suction pipe. A foot valve which is a non – return valve or one – way type of valve is fitted at the lower end of the suction pipe. The foot valve opens only in the upward direction. A strainer is also fitted at the lower end of the suction pipe.

4. Delivery pipe

A pipe whose one end is connected to the outlet of the pump and other end delivers the water at a required height is known as delivery pipe.

Work done by the Impeller

The expression of the work done by the impeller of a centrifugal pump on the liquid flowing through it may be derived in the same way as for a turbine. The liquid enters the impeller at its centre and leaves at its periphery. Fig.6.2 shows a portion of the impeller of a centrifugal pump with one vane and the velocity triangles at the inlet and outlet tips of the vane. V is absolute velocity of liquid, u is tangential velocity of the impeller, V_r is relative velocity of liquid, V_f is velocity of flow of liquid, and V_w is velocity of whirl of the liquid at the entrance to the impeller. Similarly V_1 , u_1 , V_{r1} , V_{f1} and V_{w1} represent their counterparts at the exit point of the impeller.

Velocity triangle of an impeller vane

θ = the impeller vane angle at the entrance

ϕ = the impeller vane angle at the outlet

α = the angle between the directions of the absolute velocity of entering liquid and the peripheral velocity of the impeller at the entrance

β = the angle between the absolute velocity of leaving liquid and the peripheral velocity of the impeller at the exit point

Work done per second by the impeller on the liquid may be written as

$$\text{Work done} = \frac{W}{g} (V_{w1} u_1 - V_w u)$$

Where W kg of liquid per second passes through the impeller since the liquid enters the

impeller radially $\alpha = 90^\circ$ and hence $V_w = 0$. Thus equation (6.1) becomes

$$\text{Work done} = \frac{W}{g} (V_{w1} u_1)$$

Performance of Pumps- Characteristic Curves

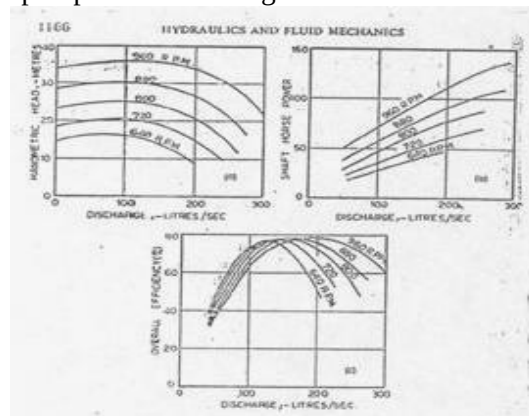
A pump is usually designed for one speed, flow rate and head in actual practice, the operation may be at some other condition of head on flow rate, and for the changed conditions, the behaviour of the pump may be quite different. Therefore, in order to predict the behaviour and performance of a pump under varying conditions, tests are performed and the results of the tests are plotted. The curves thus obtained are known as the characteristic curves of the pump. The following three types of characteristic curves are usually prepared for the centrifugal pumps:

- Main and operating characteristics.
- Constant efficiency or Muschel curves.
- Constant head and constant discharge curves.

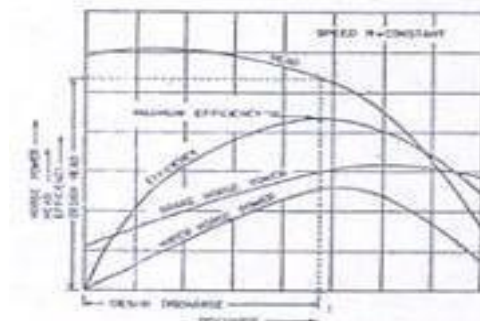
Main and Operating Characteristics

In order to obtain the main characteristic curves of a pump it is operated at different speeds. For each speed the rate of flow Q is varied by means of a delivery valve and for the different values of Q the corresponding values of manometric head H_m , shaft H.P., P , and overall efficiency η are measured or calculated. The same operation is repeated for different speeds of the pump. Then Q v/s H_m ; Q v/s P and Q v/s η curves for different speeds are plotted, so that three sets of curves, as shown in Fig.6.4 are obtained, which represent the *main characteristics* of a pump. The main characteristics are useful in indicating the performance of a pump at different speeds.

During operation a pump is normally required to run at a constant speed, which is its designed speed, (same as the speed of the driving motor). As such that particular set of main characteristics which corresponds to the designed speed is mostly used in the operations of a pump and is, therefore, known as the *operating characteristics*. A typical set of such characteristics of a pump is shown in Fig.



Main characteristics of centrifugal pump



Operating characteristics of centrifugal pump

Reciprocating Pump

The pump is the hydraulic machines which convert the mechanical energy into hydraulic energy, which is mainly in the form of pressure energy. If the mechanical energy is converted into hydraulic energy, by means of centrifugal force acting on the liquid, the pump is known as centrifugal pump. But if the mechanical energy is converted into hydraulic energy (or pressure energy) by sucking the liquid into a cylinder in which a piston is reciprocating (moving backwards and forwards), which exerts the thrust on the liquid and increases its hydraulic energy (pressure energy), the pump is known as reciprocating pump.

Classification of Reciprocation Pumps

The reciprocating pumps may be classified as:

1. According to the water being in contact with one side or both sides of the piston
2. According to the number of cylinders provided.

If the water is in contact with one side of the piston, the pump is known as single – acting. On the other hand, if the water is in contact with both sides of the piston, the pump is called double – acting. Hence, classification according to the contact of water is:

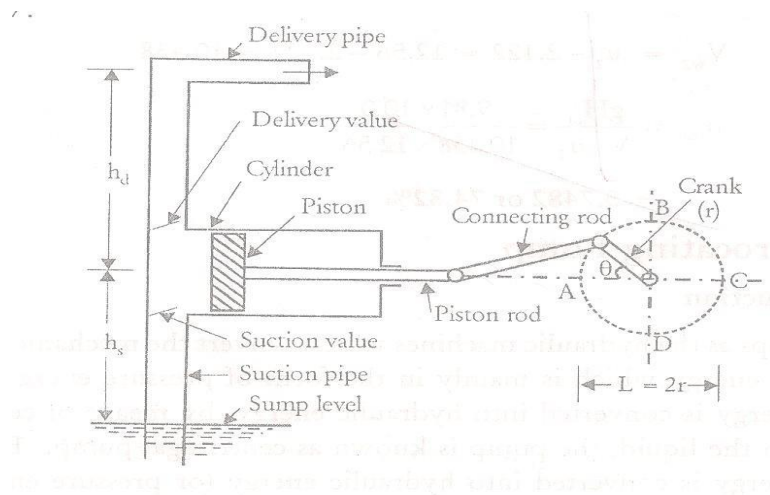
- i) Single- acting pump
- ii) Double – acting pump.

According to the number of cylinder provided, the pumps are classified as:

- i) Single cylinder pump
- ii) Double cylinder pump
- iii) Triple cylinder pump

Working Principle of a Reciprocating Pump

The following are the main parts of a reciprocating pump.



1. A cylinder with a piston, piston rod, connecting rod and a crank
2. Suction pipe
3. Delivery pipe
4. Suction valve
5. Delivery valve

Shows a single action reciprocation pump, which consists of a piston which moves forwards and backwards in a close fitting cylinder. The movement of the piston is connecting the piston rod to crank by means of a connecting rod. The crank is rotated by means of an electric motor. Suction and delivery pipes with suction valve and delivery valve are connected to the cylinder. The suction and delivery valves are one way valves or non – return valves, which allow the water to flow in one direction only. Suction valve allows water from suction pipe to the cylinder which delivery valve allows water from cylinder to delivery pipe only.

When crank starts rotating, the piston moves top and pro in the cylinder. When crank is at A, the piston is at the extreme left position in the cylinder. As the crank is rotating from A to C, (i.e., from $\theta = 0$ to $\theta = 180$ degree), the piston is moving towards right in the cylinder. The movement of the piston towards right creates a partial vacuum in the cylinder. But on the surface of the liquid in the sump atmospheric pressure is acting, which is more than the pressure inside the cylinder. Thus the liquid if forced in the suction pipe from the sump. This liquid opens the suction valve and enters the cylinder.

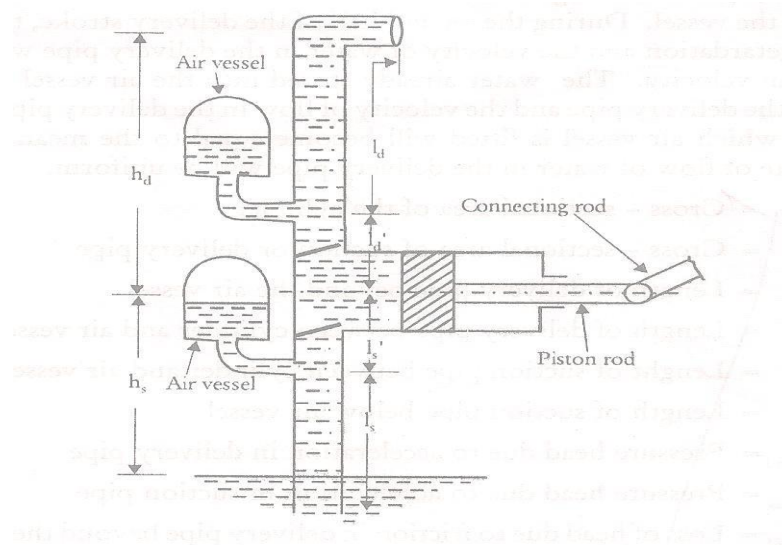
When crank is rotating from C to A (i.e., from $\theta = 180$ degree to $\theta = 360$ degree), the piston from its extreme right position starts moving towards left in the cylinder. The movement of the piston towards left increases the pressure of the liquid inside the cylinder more than atmospheric pressure. Hence suction valve closes and delivery valve closes and delivery valve opens. The liquid is forced into the delivery pipe and is raised to a required height.

Air vessels

An air vessel is a closed chamber containing compressed air in the top portion and liquid (or water) at the bottom of the chamber. At the base of the chamber there is an opening through which the liquid (or water) may flow into the vessel or out from the vessel. When the liquid enters the air vessel, the air gets compressed further and when the liquid flows out the vessel, the air will expand in the chamber.

An air vessel is fitted to the suction pipe and to the delivery pipe at a point close to the cylinder of a single – acting reciprocating pump:

- i) To obtain a continuous supply of liquid at a uniform rate
- ii) To save a considerable amount of work in overcoming the frictional Resistance in the suction and delivery pipes
- iii) To run the pump at a high speed without separation.



The single acting reciprocating to which air vessels are fitted to the suction and delivery pipes. The air vessels act like an intermediate reservoir. During the first half of the suction stroke, the piston moves with acceleration, which means the velocity of water in the suction pipe is more than the mean velocity and hence the discharge of water entering the cylinder will be more than the mean discharge. This excess quantity of water will be supplied from the air vessel to the cylinder in such a way that the velocity in the suction pipe below the air vessel is equal to mean velocity of flow. During the second half of the suction pipe is less than the mean velocity of flow. Thus the discharge entering the cylinder will be less than the mean discharge. The velocity of water in the suction pipe due to air vessel is equal to mean velocity of flow and discharge required in cylinder is less than the mean discharge. Thus the excess water flowing in suction pipe will be stored in to air vessel, which will be supplied during the first half of the next suction stroke.

When the air vessel if fitted to the delivery pipe, during the first half of delivery stroke the piston moves with acceleration and forces the water into the delivery pipe with a velocity more than the mean velocity. The quantity of water in excess of the mean discharge will flow into the air vessel. This will compress the air inside the vessel. During the second half of the delivery stroke, the piston moves with retardation and the velocity of water in the delivery pipe will be less than the mean velocity. The water already stored into the air vessel will start flowing into the delivery pipe and the velocity of flow in the delivery pipe beyond the point to which air vessel is fitted will become equal to the mean velocity. Hence the rate of flow of water in the delivery pipe will be uniform.

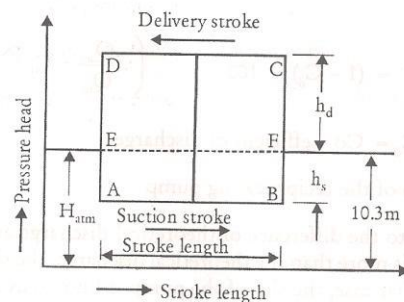
Ideal and Actual Indicator Diagram

The indicator diagram for a reciprocating pump is defined as the graph between the pressure head in the cylinder and the distance traveled by piston from inner dead centre for one complete revolution of the crank. As the maximum distance traveled by the piston is equal to the stroke length and hence the indicator diagram is a graph between pressure head and stroke length of the piston for one complete revolution. The pressure head is taken as ordinate and stroke length as abscissa.

Ideal indicator diagram

The graph between pressure head in the cylinder and stroke length of the piston for one complete revolution of the crank under ideal conditions is known as ideal indicator diagram. Shows the ideal indicator diagram, in which line EF represents the atmospheric pressure head equal to 10.3 of water.

Let H_{atm} = Atmospheric pressure head = 10.3 m of water
 L = Length of the stroke
 h_s = Suction head, and
 h_d = Delivery head.



During suction stroke, the pressure head in the cylinder is constant and equal to suction (h_s), which is below the atmospheric pressure head (H_{atm}) by a height of h_s .

The pressure head during suction stroke is represented by a horizontal line AB which is below the line EF by a height of ' h_s '.

During delivery stroke, the pressure head in the cylinder is constant and equal to delivery head (h_d), which is above the atmospheric head by a height of (h_d). Thus, the pressure head during delivery stroke is represented by a horizontal line CD which is above the line EF by a height of h_d . thus, for one complete revolution of the crank, the pressure head in the cylinder is represented by the diagram A – B – C – D.

Now, we know that the work done by the pump per second

$$= p \times g \times ALN / 60 \times (h_s + h_d)$$

$$= K \times L (h_s + h_d)$$

[where $K = pgAN / 60 = \text{Constant}$]

$$= \infty L \times (h_s + h_d) \quad \dots (i)$$

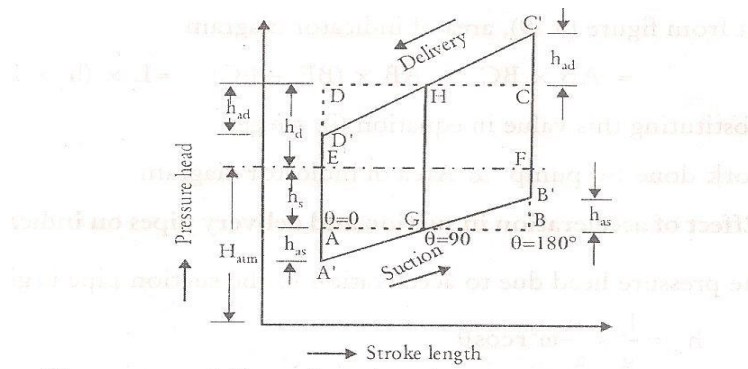
But from figure, area of indicator diagram

$$= AB \times BC = AB \times (BF + FC) = L \times (h_s + h_d)$$

Substituting this value in equation (i), we get

Work done by pump $\propto$ Area of indicator diagram

Effect of acceleration in suction and delivery pipes on indicator diagram



The pressure head due to acceleration in the suction pipe is given by

$$h_{as} = \frac{1_s}{g} \times \frac{A}{a_s} \omega^2 r \cos \theta$$

When $\theta = 0^\circ$, $\cos \theta = 1$, and $h_{as} = \frac{1_s}{g} \times \frac{A}{a_s} \omega^2 r$

When $\theta = 90^\circ$, $\cos \theta = 0$, and $h_{as} = 0$

When $\theta = 180^\circ$, $\cos \theta = -1$ and $h_{as} = -\frac{1_s}{g} \times \frac{A}{a_s} \omega^2 r$

Thus, the pressure head inside the cylinder during suction stroke will not be equal to ' h_s ' as was the case for ideal indicator diagram, but it will be equal to the sum of ' h_s ' and ' h_{as} '. As the beginning of suction stroke $\theta = 0^\circ$, ' h_{as} ' is +ve and hence the pressure head in the cylinder will be $(h_s + h_d)$ below the atmospheric pressure head. At the middle of suction stroke $\theta = 90^\circ$ and $h_{as} = 0$ and hence pressure head in the cylinder will be h_s below the atmospheric pressure head. At the end of suction stroke, $\theta = 180^\circ$ and h_{as} is -ve and hence the pressure head in the cylinder will be $(h_s - h_d)$ below the atmospheric pressure head. For suction stroke, the indicator diagram will be shown by A 'GB'. Also the area of A' AG = Area of BGB'.

Similarly, the indicator diagram for the delivery stroke can be drawn. At the beginning of delivery stroke, h_{ad} is +ve and hence the pressure head in the cylinder will be $(h_d + h_{ad})$ above the atmospheric pressure head. At the middle of the delivery stroke, $h_{ad} = 0$ and hence pressure head in the cylinder is equal h_d above the atmospheric pressure head. At the end of the delivery stroke, h_d is -ve and hence pressure in the cylinder will be $(h_d + h_{ad})$ above the atmospheric pressure head. And thus the indicator diagram for delivery stroke is represented by the line C'HD'. Also the area of CC'H = Area of DD'H.

From figure, it is now clear that due to acceleration in suction and delivery pipe, the indicator diagram has changed from ABCD to A'B'C'D'. But the area of indicator diagram ABCD = Area A'B'C'D'. Now from equation (s.22), work done, by pump is proportional to the area of indicator diagram. Hence the work done by the pump on the water remains same.

CENTRIFUGAL PUMP

1. The internal and external diameters of the centrifugal pump are 200mm and 400mm respectively. The pump is running at 1200 r.p.m. The vane angles of the impeller at inlet and outlet are 20° and 30° respectively. The water enters the impeller radially and velocity of flow is constant. Determine the work done by the impeller per unit weight of water.

Given:

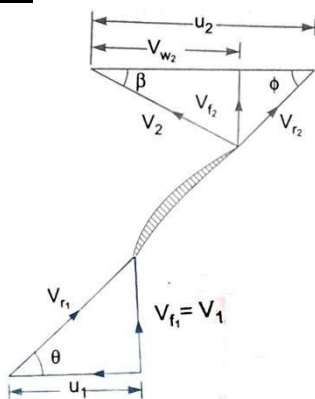
$$\begin{aligned} D_1 &= 0.2 \text{ m} \\ D_2 &= 0.4 \text{ m} \\ N &= 1200 \text{ r.p.m} \\ &= 20^\circ \\ &= 30^\circ \end{aligned}$$

$$V_{f1} = V_{f2}$$

To Find:

W.D per unit weight = ?

Solution:



$$u_1 = \text{---} = 12.56 \text{ m/s}$$

$$u_2 = \text{---} = 25.13 \text{ m/s}$$

From Inlet Velocity triangle

$$= \text{---}$$

$$V_{f1} = 4.57 \text{ m/s}$$

From outlet Velocity triangle

$$V_{f1} = V_{f2} = 4.57 \text{ m/s}$$

$$= \text{---}$$

$$V_{w2} = 17.215 \text{ m/s}$$

Work done per unit weight:

$$\text{W.D} = \text{---}$$

$$\text{W.D} = 44.1 \text{ Nm/N}$$

2. A centrifugal pump is to be discharge $0.118 \text{ m}^3/\text{s}$ at a speed of 1450 r.p.m. against a head of 25 m. The impeller diameter is 250mm, its width at outlet is 50mm and manometric efficiency is 75 %. Determine the vane angle at outer periphery of the impeller.

Given:

$$\begin{aligned} Q &= 0.118 \text{ m}^3/\text{s} \\ N &= 1450 \text{ r.p.m} \\ H_m &= 25 \text{ m} \\ D_2 &= 0.25 \text{ m} \\ B_2 &= 0.05 \text{ m} \\ \eta_{\text{man}} &= 0.75 \end{aligned}$$

To find:

$$=?$$

Solution:

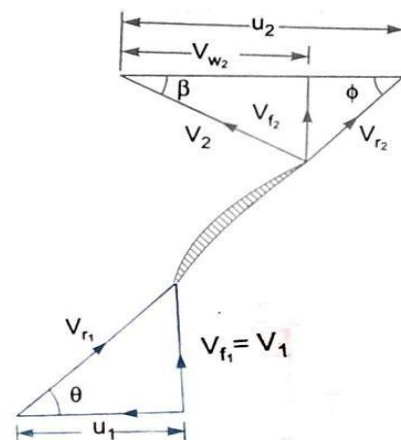
$$u_2 = \text{---} = 18.98 \text{ m/s}$$

$$= \text{---}_{\text{man}}$$

$$V_{w2} = 17.23 \text{ m/s}$$

$$Q = D_2 B_2 V_{f2}$$

$$V_{f2} = 3.0 \text{ m/s}$$



$$= \text{---}$$

$$= 59^\circ 44'$$

3. A centrifugal pump delivers water against a net head of 14.5 m and a design speed of 1000 r.p.m. The vanes are curved back to an angle of 30° with the periphery. The impeller diameter is 300 mm and outlet width is 50mm. determine the discharge of the pump if manometric efficiency is 95 %.

Given:

$$\begin{aligned} H_m &= 14.5 \text{ m} \\ N &= 1000 \text{ r.p.m} \\ &= 30^\circ \\ D_2 &= 0.3 \text{ m} \\ B_2 &= 0.05 \text{ m} \\ \eta_{\text{man}} &= 0.95 \end{aligned}$$

To Find:

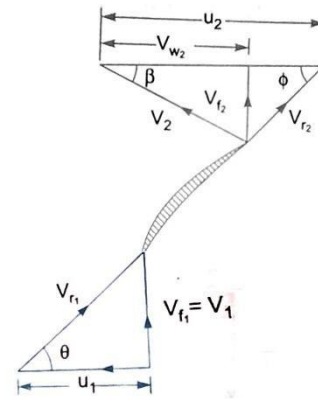
$$Q = ?$$

Solution:

$$u_2 = \frac{\pi D_2 N}{60} = 15.70 \text{ m/s}$$

$$\eta_{\text{man}} = \frac{H_m}{H_{\text{theoretical}}}$$

$$V_{w2} = 9.54 \text{ m/s}$$



$$= \frac{H_m}{H_{\text{theoretical}}}$$

$$V_{f2} = 3.556 \text{ m/s}$$

$$Q = D_2 B_2 V_{f2}$$

$$Q = 0.1675 \text{ m}^3/\text{s}$$

4. A centrifugal pump having outer diameter equal to two times the inner diameter and running at 1000 r.p.m works against a total head of 40m. The velocity of flow through the impeller is constant and equal to 2.5 m/s. the vanes are set back at an angle of 40° at outlet. If the outer diameter of the impeller is 500mm and width at outlet is 50mm, determine: (i) vane angle at inlet (ii) work done by the impeller on water per second and (iii) manometric efficiency.

Given:

$$\begin{aligned} D_2 &= 2 \times D_1 \\ N &= 1000 \text{ r.p.m} \\ H_m &= 40 \text{ m} \\ V_{f1} = V_{f2} &= 2.5 \text{ m/s} \\ &= 40^\circ \\ D_2 &= 0.5 \text{ m} \\ B_2 &= 0.05 \text{ m} \end{aligned}$$

$$u_2 = \frac{\pi D_2 N}{60} = 26.18 \text{ m/s}$$

$$Q = D_2 B_2 V_{f2} = 0.1963 \text{ m}^3/\text{s}$$

From Inlet Velocity triangle

$$= \frac{H_m}{H_{\text{theoretical}}}$$

$$= 10^\circ 48'$$

From outlet Velocity triangle

$$V_{f1} = V_{f2} = 2.5 \text{ m/s}$$

$$= \frac{H_m}{H_{\text{theoretical}}}$$

$$V_{w2} = 23.2 \text{ m/s}$$

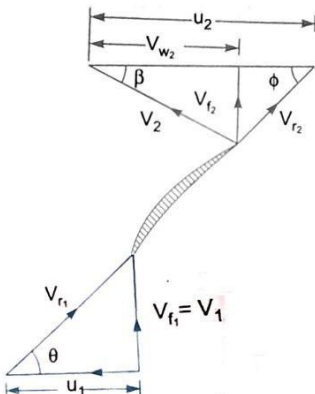
Work done per second:

$$\text{W.D per second} = Q V_{w2} u_2 = 119227.9 \text{ Nm/s}$$

Manometric efficiency:

$$\eta_{\text{man}} = \frac{H_m}{H_{\text{theoretical}}} = 64.4 \%$$

Solution:



$$D_1 = 0.25 \text{ m}$$

$$u_1 = \frac{\pi D_1 N}{60} = 13.09 \text{ m/s}$$

5. A centrifugal pump discharge $0.15 \text{ m}^3/\text{s}$ of water against a head of 12.5 m, the speed of the impeller being 600 r.p.m. The outer and inner diameters of impeller are 500 mm and 250 mm respectively and the vanes are bent back at 35° the tangent at exit. If the area of flow remains 0.07 m^2 from inlet to outlet, calculate: (i) Manometric efficiency of the pump (ii) vane angle at inlet (iii) loss of head at inlet to impeller when the discharge is reduced by 40% without changing the speed.

Given:

$$\begin{aligned} Q &= 0.15 \text{ m}^3/\text{s} \\ H_m &= 12.5 \text{ m} \\ N &= 600 \text{ r.p.m} \\ D_2 &= 0.5 \text{ m} \quad D_1 \\ &= 0.25 \text{ m} \\ &= 35^\circ \\ A_1 &= A_2 = 0.07 \text{ m}^2 \end{aligned}$$

To find:

$$\eta_{\text{man}} = ?$$

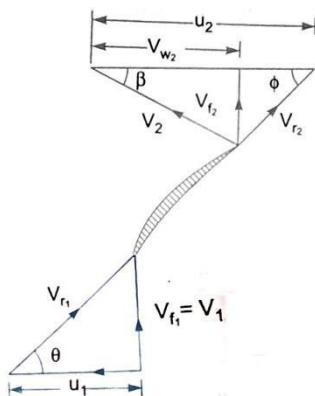
$$= ?$$

Loss of head at inlet when discharge is reduced by 40% = ?

Solution:

$$\begin{aligned} Q &= A_1 V_{f1} \\ V_{f1} &= V_{f2} = 2.14 \text{ m/s} \\ u_1 &= 7.85 \text{ m/s} \end{aligned}$$

$$u_2 = 15.70 \text{ m/s}$$



From outlet Velocity triangle

$$= \frac{V_{f2}}{V_2}$$

$$V_{w2} = 12.64 \text{ m/s}$$

a. Manometric efficiency:

$$\eta_{\text{man}} = \frac{H_m}{H_{\text{theoretical}}} = 61.8\%$$

b. Vane angle at inlet:

From Inlet Velocity triangle

$$= \frac{V_{f1}}{u_1}$$

$$= 15^\circ 12'$$

c. Loss of head at inlet when discharge is reduced by 40%

Discharge is reduced by 40%. Hence the new discharge is given by

$$Q^* = 0.6 Q = 0.09 \text{ m}^3/\text{s}$$

$$Q^* = A \times V_{f1}^*$$

$$V_{f1}^* = 1.284 \text{ m/s}$$

$$= \frac{Q^*}{A}$$

$$u_1^* = 4.808 \text{ m/s}$$

$$\text{Head loss at inlet} = \frac{u_1^2 - u_1^{*2}}{2g} = 0.5 \text{ m}$$

6. The outer diameter of an impeller of the centrifugal pump is 400 mm and outlet widths is 50mm. the pump is running at 800 r.p.m. and working against a total head of 15m. The vanes angle at outlet is 40° and manometric efficiency is 75 %. Determine: (i) velocity of flow at outlet (ii) velocity of water leaving the vane (iii) angle made by the absolute velocity at outlet with the direction of motion at outlet (iv) Discharge

Given:

$$\begin{aligned} D_2 &= 0.4 \text{ m} \\ B_2 &= 0.05 \text{ m} \\ N &= 800 \text{ r.p.m} \\ H_m &= 15 \text{ m} \\ &= 40^\circ \end{aligned}$$

$$\eta_{\text{man}} = 0.75$$

To Find:

$$V_{f2} = ?$$

$$V_2 = ?$$

$$= ?$$

$$Q = ?$$

Solution:

From outlet Velocity triangle

$$u_2 = \frac{V_{f2}}{\sin \phi} = 16.75 \text{ m/s}$$

$$= \frac{4.23}{\sin 15.48^\circ}$$

$$\eta_{\text{man}} = \frac{V_{f2}}{V_2} = \frac{4.23}{12.45} = 0.34$$

$$V_{f2} = 4.23 \text{ m/s}$$

$$= 15.48^\circ$$

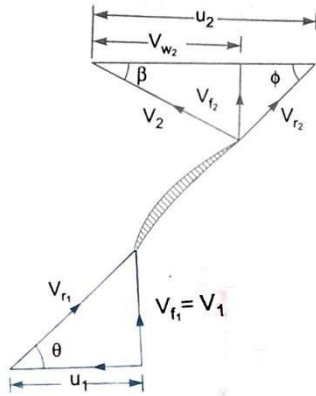
$$V_{w2} = 11.71 \text{ m/s}$$

$$= 19^\circ 48'$$

$$= 12.45 \text{ m/s}$$

$$V_2 = 12.45 \text{ m/s}$$

$$Q = D_2 B_2 V_{f2} = 0.265 \text{ m}^3/\text{s}$$



7. A centrifugal pump is running at 1000 r.p.m. The outlet vane angle of the impeller is 45° and velocity of flow at outlet is 2.5 m/s. The discharge through the pump is 200 liters/s where the pump working against a total head of 20 m. if the manometric efficiency of the pump is 80%. Determine (i) the diameter of the impeller and (ii) the width of the impeller at outlet.

Given:

$$N = 1000 \text{ r.p.m}$$

$$\eta_{\text{man}} = 0.80$$

$$\phi = 45^\circ$$

$$V_{w2} \times u_2 = 245.25 \text{ ----- (1)}$$

$$V_{f2} = 2.5 \text{ m/s}$$

$$= \frac{2.5}{\sin 45^\circ}$$

$$Q = 200 \text{ liters/s} = 0.2 \text{ m}^3/\text{s}$$

$$u_2 - V_{w2} = 2.5 \text{ ----- (2)}$$

$$H_m = 20 \text{ m}$$

Substitute equation (1) in (2)

$$\eta_{\text{man}} = 0.80$$

$$u_2 = 16.96 \text{ or } -14.46 \text{ ----- (Negative value not possible)}$$

To find:

$$D_2 = ?$$

$$u_2 = \frac{2\pi N D_2}{60}$$

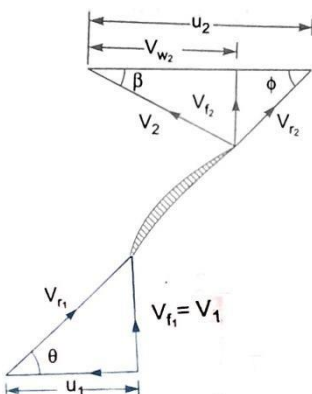
$$B_2 = ?$$

$$D_2 = 0.324 \text{ m}$$

Solution:

$$Q = D_2 B_2 V_{f2}$$

$$B_2 = 0.0786 \text{ m}$$



8. A centrifugal pump has a following dimension: inlet radius = 80mm; outlet radius = 160mm; width of the impeller at inlet = 50 mm; $\beta_1 = 0.45$ radians; $\beta_2 = 0.25$ radians; width of the impeller at outlet = 50 mm. assuming shock less entry determine the discharge and the head developed by the pump when the impeller rotates at 90radians/second.

Given:

$$R_1 = 0.08 \text{ m}$$

$$R_2 = 0.16 \text{ m}$$

$$B_1 = 0.05 \text{ m}$$

$$B_2 = 0.05 \text{ m}$$

$$\beta_1 = 0.45 \text{ radians} = 0.45 \times \frac{180}{\pi} = 25.78^\circ$$

$$\beta_2 = 0.25 \text{ radians} = 0.25 \times \frac{180}{\pi} = 14.32^\circ$$

$$\omega = 90 \text{ rad/s}$$

$$u_1 = \omega R_1 = 7.2 \text{ m/s}$$

$$u_2 = \omega R_2 = 14.4 \text{ m/s}$$

From Inlet Velocity triangle

$$= \text{---}$$

$$V_{f1} = 3.478 \text{ m/s}$$

a. Discharge:

$$Q = D_1 B_1 V_{f1} = 0.0874 \text{ m}^3/\text{s}$$

To find:

Discharge (Q) =?

Head (H_m) =?

b. Head Developed:

$$Q = D_2 B_2 V_{f2}$$

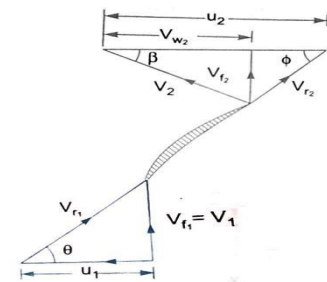
$$V_{f2} = 1.7387 \text{ m/s}$$

$$= \text{---}$$

$$V_{w2} = 7.951 \text{ m/s}$$

$$H_m = \text{---}$$

$$H_m = 11.142 \text{ m}$$



9. The internal and external diameter of the centrifugal pump which is running at 1000 r.p.m. is 200mm and 400mm respectively. The discharge through the pump is $0.04 \text{ m}^3/\text{s}$ and velocity of flow is constant and equal to 2.0 m/s . the diameter of the suction and delivery pipes are 150mm and 100mm respectively and suction and delivery heads are 6 m and 30m of water respectively. If the outlet vane angle is 45° and the power required to drive the pump is 16.186 kW, determine: (i) the vane angle of the impeller at inlet (ii) the overall efficiency of the pump (iii) manometric efficiency of the pump.

Given:

$$N = 1000 \text{ r.p.m}$$

$$D_1 = 0.2 \text{ m}$$

$$D_2 = 0.4 \text{ m}$$

$$Q = 0.04 \text{ m}^3/\text{s}$$

$$V_{f1} = V_{f2} = 2.0 \text{ m/s}$$

$$D_s = 0.15 \text{ m}$$

$$D_d = 0.10 \text{ m}$$

$$h_s = 6 \text{ m}$$

$$h_d = 30 \text{ m}$$

$$\beta_2 = 45^\circ$$

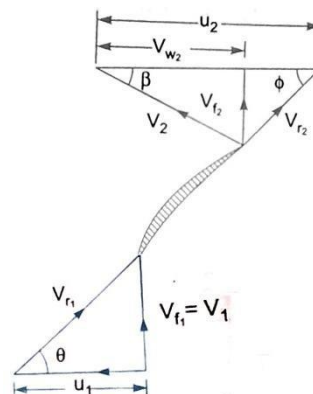
$$P = 16.186 \text{ kW}$$

To Find:

$$\beta_1 = ?$$

$$\eta_o = ?$$

$$\eta_{man} = ?$$



$$u_1 = \text{---} = 10.47 \text{ m/s}$$

$$u_2 = \text{---} = 20.94 \text{ m/s}$$

a. From Inlet Velocity triangle

Solution:

$$= \text{---}$$

$$= 10^\circ 48'$$

b. Overall efficiency

$$Q = A_S V_S = A_d V_d$$

$$A_S = \pi D_S^2 = 0.01767 \text{ m}^2$$

$$A_d = \pi D_d^2 = 0.007853 \text{ m}^2$$

$$V_S = 2.26 \text{ m/s}$$

$$V_d = 5.09 \text{ m/s}$$

$$H = h_m + h_s + h_d + \text{---} - \text{---}$$

$$H_m = 25.06 \text{ m}$$

$$\text{Water power} = \text{---}, \text{ where,}$$

$$W = \rho g Q H = 1000 \times 9.81$$

$$\text{W.P} = 9.833 \text{ kW}$$

$$\eta_o = \frac{\text{---}}{\text{---}} = 60.74 \%$$

c. Manometric efficiency:

$$= \frac{\text{---}}{\text{---}}$$

$$V_{w2} = 18.94 \text{ m/s}$$

$$\eta_{\text{man}} = \frac{\text{---}}{\text{---}} = 61.98 \%$$

10. Find the power required to drive a centrifugal pump which delivers $0.04 \text{ m}^3/\text{s}$ of water to a height of 20m through a 15 cm diameter pipe and 100m long. The overall efficiency of the pump is 70% and co-efficient of friction 'f' = 0.15 in the formula $h_f = \text{---}$

Given:

$$Q = 0.04 \text{ m}^3/\text{s}$$

$$H_S = h_s + h_d = 20 \text{ m}$$

$$D = D_S = D_d = 0.15 \text{ m}$$

$$L = L_S + L_d = 100 \text{ m}$$

$$\eta_o = 0.70$$

$$f = 0.15$$

$$h_f = h_{fs} + h_{fd} = \text{---}$$

$$= 10.41 \text{ m}$$

$$H_m = h_s + h_d + h_{fs} + h_{fd} + \text{---}$$

$$= 30.67 \text{ m}$$

$$\text{Water power} = \text{---}, \text{ where,}$$

$$W = \rho g Q H = 1000 \times 9.81$$

$$\text{W.P} = 12.03 \text{ kW}$$

$$\eta_o = \frac{\text{---}}{\text{---}}$$

$$\text{S.P} = 17.19 \text{ kW}$$

To Find:

$$\text{Shaft Power} = ?$$

Solution:

$$Q = A_S V_S = A_d V_d$$

$$V_S = V_d = 2.26 \text{ m/s}$$

11. Find the rise in pressure in the impeller of a centrifugal pump through which water is flowing at a rate of $0.01 \text{ m}^3/\text{s}$. the internal and external diameters of the impeller are 15cm and 30cm respectively. The widths of the impeller at inlet and outlet are 1.2cm and 0.6cm. The pump is running at 1500 r.p.m. the water enters the impeller radially at inlet and impeller vane angle at outlet is 45° . Neglect losses through the impeller.

Given:

$$Q = 0.01 \text{ m}^3/\text{s}$$

$$D_1 = 0.15 \text{ m}$$

$$D_2 = 0.30 \text{ m}$$

$$B_1 = 0.012 \text{ m}$$

$$B_2 = 0.006 \text{ m}$$

$$N = 1500 \text{ r.p.m}$$

To Find:

$$\text{Pressure Raise} = ?$$

Solution:

$$Q = D_1 B_1 V_{f1}$$

$$V_{f1} = 1.768 \text{ m/s}$$

$$= 45^\circ$$

$$Q = D_2 B_2 V_{f2}$$

$$V_{f2} = 1.768 \text{ m/s}$$

$$u_2 = \text{---} = 23.56 \text{ m/s}$$

$$\text{Pressure raise} = \text{---} [V_{f1}^2 + u_2^2 - V_{f2}^2 \text{cosec}^2]$$

$$= 28.13 \text{ m}$$

12. The diameters of an impeller of a centrifugal pump at inlet and outlet are 30cm and 60cm respectively. Determine the minimum starting speed of the pump if its works against a head of 30m.

Given:

$$D_1 = 0.30 \text{ m}$$

$$D_2 = 0.60 \text{ m}$$

$$H_m = 30 \text{ m}$$

$$u_1 = \text{---} = 0.0157 \text{ N}$$

$$u_2 = \text{---} = 0.03141 \text{ N}$$

Equations for minimum speed

$$\text{---} - \text{---} = H_m$$

Minimum starting speed (N) =?

$$N = 891.8 \text{ r.p.m}$$

To Find:

Solution:

13. The diameters of an impeller of a centrifugal pump at inlet and outlet are 30cm and 60cm respectively. The velocity of flow at outlet is 2.0 m/s and the vanes are set back at an angle of 45° at outlet. Determine the minimum starting speed of the pump if the manometric efficiency is 70 %.

Given:

$$D_1 = 0.30 \text{ m}$$

$$D_2 = 0.60 \text{ m}$$

$$V_{f2} = 2.0 \text{ m/s}$$

$$= 45^\circ$$

$$\eta_{man} = 0.70$$

$$= \text{---}$$

$$V_{w2} = u_2 - 2 = 0.03141 \text{ N} - 2 \text{ ----- (1)}$$

$$= \text{---}_{man}$$

Minimum Starting speed (N) =?

$$H_m = 0.07135(0.03141 \text{ N} - 2) 0.03141 \text{ N}$$

Equations for minimum speed

$$\text{---} - \text{---} = H_m$$

$$u_1 = \text{---} = 0.0157 \text{ N}$$

$$u_2 = \text{---} = 0.03141 \text{ N}$$

$$N = 137.22 \text{ r.p.m}$$

Solution:

14. A centrifugal pump with 1.2 diameter runs at 200 r.p.m and pumps 1880 liters/s, the average lift being 6m. The angle which the vane makes at exit with tangent to the impeller is 26° and the radial velocity of flow is 2.5 m/s. determine the manometric efficiency and the least speed to start pumping against a head of 6m, the inner diameter of the impeller being 0.6m.

Given:

$$D_2 = 1.2 \text{ m}$$

$$N = 200 \text{ r.p.m}$$

$$Q = 1880 \text{ liters/s} = 1.88 \text{ m}^3/\text{s}$$

$$H_m = 6 \text{ m}$$

$$V_{f2} = 2.5 \text{ m/s}$$

$$= 26^\circ$$

$$D_1 = 0.6 \text{ m}$$

To Find:

Manometric efficiency =?

Minimum Speed =?

Solution:

a. Manometric efficiency:

$$u_2 = \text{---} = 12.56 \text{ m/s}$$

$$= \frac{\quad}{\quad}$$

$$V_{w2} = 7.43 \text{ m/s}$$

$$\eta_{\text{man}} = \frac{\quad}{\quad}$$

$$= 63 \%$$

$$u_1 = \frac{\quad}{\quad} = 0.03141 \text{ N}$$

$$u_2 = \frac{\quad}{\quad} = 0.06283 \text{ N}$$

$$\frac{\quad}{\quad} = H_m$$

$$N = 200 \text{ r.p.m}$$

b. Least speed to start the pump:

15. A three stage centrifugal pump has impeller 40cm in diameter and 2cm wide at outlet. The vanes are curved back at outlet at 45° and reduce the circumferential area by 10 %. The manometric efficiency is 90 % and the overall efficiency is 80 %. Determine the head generated by the pump when running at 1000 r.p.m. delivering 50 liters per second. What should be the shaft horse power?

Given:

$$n = 3$$

$$D_2 = 0.40 \text{ m}$$

$$B_2 = 0.02 \text{ m}$$

$$\alpha = 45^\circ$$

$$\text{Reduction in area at outlet} = 10\% = 0.1$$

$$\eta_{\text{man}} = 0.90$$

$$\eta_o = 0.80$$

$$N = 1000 \text{ r.p.m}$$

$$Q = 0.05 \text{ m}^3/\text{s}$$

To Find:

Head generated by the pump =?

Shaft Power =?

Solution:

$$\text{Area of flow at outlet} = 0.9 D_2 B_2 = 0.02262 \text{ m}^2$$

$$V_{f2} = Q/A_2 = 2.21 \text{ m/s}$$

$$u_2 = \frac{\quad}{\quad} = 20.94 \text{ m/s}$$

$$= \frac{\quad}{\quad}$$

$$V_{w2} = 18.73 \text{ m/s}$$

a. Head generated by the pump:

$$= \frac{\quad}{\eta_{\text{man}}}$$

$$H_m = 35.98 \text{ m}$$

$$\text{Total head generated by the pump} = n \times H_m$$

$$= 3 \times 35.98 = 107.94 \text{ m}$$

b. Shaft Power:

Water power = $\rho g Q H_m$, where,

$$W = \rho g Q H_m = 1000 \times 9.81$$

$$W.P = 52.94 \text{ kW}$$

$$\eta_o = \frac{\quad}{\quad}$$

$$S.P = 66.175 \text{ kW}$$

16. A four stage centrifugal pump has four identical impellers, keep to the same shaft. The shaft running at 400 r.p.m. and the total manometric head developed by the multistage pump is 40m. The discharge through the pump is $0.2 \text{ m}^3/\text{s}$. the vanes of each impeller having outlet angle as 45° . If the width and diameter of each impeller at outlet is 5cm and 60cm respectively, find the manometric efficiency.

Given:

$$n = 4$$

$$N = 400 \text{ r.p.m}$$

$$\text{Total head} = 40 \text{ m}$$

$$\text{For each stage } H_m = 10 \text{ m}$$

$$Q = 0.2 \text{ m}^3/\text{s}$$

$$\alpha = 45^\circ$$

$$D_2 = 0.60 \text{ m}$$

$$B_2 = 0.05 \text{ m}$$

To Find:

$$u_2 = \frac{\quad}{\quad} = 12.56 \text{ m/s}$$

$$Q = D_2 B_2 V_{f2}$$

$$V_{f2} = 2.122 \text{ m/s}$$

$$= \frac{\quad}{\quad}$$

$$V_{w2} = 10.438 \text{ m/s}$$

$$\eta_{\text{man}} = \frac{\quad}{\quad}$$

$$= 74.82 \%$$